Answer Key & Teacher Notes

Session 1 — Area of Parallelograms · Session 2 — Area of Rhombuses

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Area of Parallelograms

TALK PROMPTS

- Why must the height make a square corner with the base you chose?
- How does the area of a parallelogram compare to a rectangle with the same base and height?

PARTNER TASK

Floor Tile and Missing Height. One partner sketches and labels the parallelogram (base, height with a square corner, slanted side); the other writes A = b × h and solves. Switch roles with Paper Strip: base 16 cm, height 7 cm, slanted side 9 cm.

WATCH FOR

- Multiplying the base by the slanted side instead of the perpendicular height.
- Labeling area in linear units (cm instead of cm²).
- Multiplying instead of dividing to find a missing base or height.

SESSION 2 · TEACHER NOTES

Area of Rhombuses

TALK PROMPTS

- What do a rhombus and a parallelogram share, and what makes a rhombus special?
- Why does multiplying both full diagonals give too much area?

PARTNER TASK

Paper Kite. A rhombus has diagonals of 20 in. and 14 in. One partner cuts a paper model along the diagonals and rearranges the triangles into a rectangle; the other computes (20 × 14) ÷ 2 = 140 in². Switch roles with Quilt Patch.

WATCH FOR

- Multiplying the diagonals and forgetting to divide by 2.
- Using a slanted side length as the height in A = b × h.
- Finding a missing diagonal without doubling the area first (60 ÷ 12 = 5).

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.G.A.1	1	2	1	1	1	2	2	3	2	1	1
S1 · Form B	6.G.A.1	1	2	1	1	1	2	2	3	2	1	1
S2 · Form A	6.G.A.1	1	1	1	1	1	2	2	3	2	1	1
S2 · Form B	6.G.A.1	1	1	1	1	1	2	2	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

parallelogram: A four-sided shape whose opposite sides are **parallel**.

base: The side you **measure** the height from.

height: The distance from the base to the opposite side, at a **square** corner.

- Area of a parallelogram: A = **base** × height.
- The height makes a **square** corner with the base. The slanted side is **not** the height.
- Missing height or base: **divide** the area by the measure you know.

Example: base = **9** in., height = **4** in. Skip the 5 in. slanted side. A = 9 × 4 = **36** Answer: The area is **36** in².

YOU TRY

1. 10 × 6 = 60 cm²

2. a. 6 in b. 9 m

EXIT TICKET

1. a. 52 in² b. 9 m

2. 11 × 5 = 55 m²

3. **B** (cm²)

PRACTICE FORM A

1. a. 126 cm² b. 72 in² c. 77 ft² d. 100 m²

2. a. 12 × 5 = 60 ft² b. The 8 ft side is slanted; it does not make a square corner with the base, so it is not the height.

3. a. 15 × 10 = 150 cm² b. 3 × 150 = 450 cm²

4. 90 = 18 × h, so h = 90 ÷ 18 = 5 in.

5. 16 × 7 = 112 cm² (the 9 cm side is not used)

6. 96 = 8 × h, so h = 96 ÷ 8 = 12 in.

7. X: 10 × 13 = 130 cm². Y: 26 × 5 = 130 cm². They cover the same amount.

8. a. 15 × 8 = 120 ft² b. 120 ÷ 50 = 2.4, so 3 full cans (2 cans cover only 100 ft²) c. 30 × 8 = 240 ft², twice the first area

9. She used the slanted side. The height must make a square corner with the base: A = 20 × 15 = 300 cm².

10. **C** (90 m²)

11. **A** (9 cm)

Self-check word **SQUARES**

PRACTICE FORM B

1. a. 80 cm² b. 52 in² c. 72 ft² d. 90 m²

2. a. 14 × 6 = 84 ft² b. The 10 ft side is slanted; it does not make a square corner with the base, so it is not the height.

3. a. 12 × 9 = 108 cm² b. 3 × 108 = 324 cm²

4. 84 = 12 × h, so h = 84 ÷ 12 = 7 in.

5. 18 × 8 = 144 cm² (the 11 cm side is not used)

6. 117 = 9 × h, so h = 117 ÷ 9 = 13 in.

7. X: 9 × 14 = 126 cm². Y: 18 × 7 = 126 cm². They cover the same amount.

8. a. 20 × 6 = 120 ft² b. 120 ÷ 50 = 2.4, so 3 full cans (2 cans cover only 100 ft²) c. 40 × 6 = 240 ft², twice the first area

9. He used the slanted side. The height must make a square corner with the base: A = 14 × 8 = 112 m².

10. **C** (100 ft²)

11. **A** (9 m)

Self-check word **HEIGHTS**

SESSION 2 · ANSWERS

Area of Rhombuses

GUIDED NOTES

rhombus: A parallelogram with four sides of the **same** length.

diagonal: A segment from one corner to the **opposite** corner.

congruent: Exactly the same **size** and shape.

- A rhombus is a **parallelogram**, so A = base × height works.
- With the two diagonals, A = d1 × d2 ÷ **2**.
- Missing diagonal: **double** the area, then divide by the diagonal you know.

Example: Two diagonals are given, so use A = d1 × d2 ÷ 2. 10 × 8 = **80** 80 ÷ 2 = **40** Answer: The area is **40** cm².

YOU TRY

1. a. 36 m² b. 35 cm²

2. 2 × 72 = 144; 144 ÷ 16 = 9 in.

EXIT TICKET

1. a. 35 ft² b. 63 m²

2. (18 × 10) ÷ 2 = 90 cm²

3. **B** (12 cm)

PRACTICE FORM A

1. a. 70 in² b. 54 cm² c. 130 ft² d. 55 cm²

2. A rhombus is a parallelogram: A = 10 × 9 = 90 cm²

3. (20 × 14) ÷ 2 = 280 ÷ 2 = 140 in²

4. 84 = 12 × h, so h = 7 cm

5. 150 = b × 10, so b = 15 cm

6. (18 × 24) ÷ 2 = 432 ÷ 2 = 216 ft²

7. Double the area: 2 × 60 = 120. Divide: 120 ÷ 12 = 10 m.

8. a. (30 × 16) ÷ 2 = 240 in² b. 240 ÷ 60 = 4 strips c. (60 × 32) ÷ 2 = 960 in², 4 times as large

9. She skipped dividing by 2; 160 cm² is the rectangle around the rhombus. A = (10 × 16) ÷ 2 = 80 cm².

10. **B** (24 ft²)

11. **A** (All four sides are the same length.)

Self-check word **RHOMBUS**

PRACTICE FORM B

1. a. 54 m² b. 104 in² c. 40 m² d. 35 cm²

2. A rhombus is a parallelogram: A = 12 × 11 = 132 cm²

3. (22 × 16) ÷ 2 = 352 ÷ 2 = 176 in²

4. 126 = 14 × h, so h = 9 cm

5. 108 = b × 9, so b = 12 cm

6. (20 × 26) ÷ 2 = 520 ÷ 2 = 260 ft²

7. Double the area: 2 × 45 = 90. Divide: 90 ÷ 9 = 10 m.

8. a. (24 × 18) ÷ 2 = 216 in² b. 216 ÷ 54 = 4 strips c. (48 × 36) ÷ 2 = 864 in², 4 times as large

9. She skipped dividing by 2; 108 ft² is the rectangle around the rhombus. A = (12 × 9) ÷ 2 = 54 ft².

10. **C** (90 in²)

11. **C** ((d1 × d2) ÷ 2)

Self-check word **POLYGON**

HOMEWORK PACKET · ANSWERS

same answers for the core packet and Level 1

SHEET 1 · AREA OF PARALLELOGRAMS

1. 11 × 6 = 66 cm² (the 8 cm side is not used)

2. a. 84 ft² b. 81 m² c. 80 in² d. b = 63 ÷ 7 = 9 m

3. One space: 9 × 18 = 162 ft²; 10 spaces: 1,620 ft²

4. h = 108 ÷ 12 = 9 ft (check: 12 × 9 = 108)

5. She multiplied the base by the slanted side (10 × 7). The height must be perpendicular to the base: A = 10 × 5 = 50 cm².

SHEET 2 · AREA OF RHOMBUSES

1. (16 × 10) ÷ 2 = 80 cm²

2. a. 70 m² b. 63 ft²

3. 2 × 60 = 120; 120 ÷ 10 = 12 ft

4. A: (12 × 8) ÷ 2 = 48 ft²; B: 8 × h = 48, so h = 6 ft

5. He multiplied the diagonals but did not divide by 2. A = (18 × 8) ÷ 2 = 72 in².

6. **C** (35 m²)

7. **C** (90 in²)

October 2026 conceptual additions — teacher key

Session 1 · Area of Parallelograms

Area 8 × 3 = 24 cm². Rearranging a triangular piece preserves area and makes a rectangle with the same base and perpendicular height; the slanted side is not used.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: Area 8 × 3 = 24 cm². Rearranging a triangular piece preserves area and makes a rectangle with the same base and perpendicular height; the slanted side is not used.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Area of Rhombuses

Half-diagonals are 6 and 4; each triangle has area 12 cm²; four total 48 cm². This equals 12 × 8 ÷ 2.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: Half-diagonals are 6 and 4; each triangle has area 12 cm²; four total 48 cm². This equals 12 × 8 ÷ 2.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.